Alexander Reinthal

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reinthal

7 years of industry experience - cybersecurity operations - machine learning - data science & engineering

Publications

Data Modelling for Predicting Exploits Nov 2018
Reinthal, A, Filippakis, E., Almgren, M.
10.1007/978-3-030-03638-6_21 (Springer LNCS: Nordic Conference on Secure IT Systems)

Research Projects

Detecting Piecewise Cyber Espionage in Model APIs *Nov 2025 – Nov 2025*
Apart Research sprint (4th place out of 641 submissions) studying detection of piecewise misuse activity in model APIs

- Showed that piecewise misuse — where individual requests appear benign — can be detected by modelling activity using the cyber kill chain
- Correlated data across requests (e.g. IP addresses) to surface patterns that per-request guardrails like Llama Guard 3 miss
- Co-authored with Arthur Colle, David Williams-King, Yingquan Li, and Lihn Le

Detecting Deception in Chinese Models *Jan 2026 – Feb 2026*
ARENA 7.0 weekend hackathon (1st place) applying linear probing to detect deception and political censorship in Chinese-origin language models

- Extended "Detecting Strategic Deception Using Linear Probes" to Qwen 2.5 models, achieving AUROC 0.849 on deception detection
- Investigated whether deception and political censorship share the same linear representation in activation space using custom datasets on politically sensitive topics
- Found optimal layers for probes, conducted steering experiments, and increased deception with model size (7B to 32B), with larger models showing cleaner deception-score separation between non-sensitive and sensitive topics
- Discovered that Qwen spontaneously switches to Chinese when asked sensitive questions in English, consistent with a refusal/anti-jailbreaking mechanism

Inoculation Prompting Against Emergent Misalignment *Jan 2026 – Feb 2026*
ARENA 7.0 capstone project (3rd place) testing whether inoculation prompting during supervised fine-tuning can revert emergent misalignment in language models

- Demonstrated that inoculation prompting reduced emergent misalignment from ~12% to ~2% while maintaining model coherence (Qwen2.5-14B)
- Showed cross-task generalization: inoculation on one domain (medical/sports) reduces misalignment induced from a different domain (financial advice)
- Conducted mechanistic interpretability experiments using rank-1 LoRA adapters to test whether emergent misalignment fine-tuning shifts activations along Anthropic's "Assistant Axis"

Novel AI Control Protocol Classes: Evaluation and Scalability *Jan 2026 – present*
Ongoing AI safety research through AI Safety Camp, investigating alternative control protocol classes and their scaling properties as model capabilities increase

- Building on Greenblatt et al.'s control evaluation framework to evaluate hierarchical and parallel control structures against simple trusted-untrusted model pairs

- Researching whether complex oversight structures offer more safety-usefulness Pareto frontiers, particularly as the capability gap between trusted and untrusted models widens

Certificates

- Bluedot AGI Strategy, Completed: 2025-10-07

Experience

Alum *Jan 2026 – Feb 2026*
ARENA 7.0

- Research projects placed 1st (Detecting Deception in Chinese Models) and 3rd (Inoculation Prompting Against Emergent Misalignment).
- ARENA prepares fellows for work as researchers in technical AI Safety.
- The curriculum is tought over five high-paced weeks and teaches the transformer architecture, mechanistic interpretability, RLHF, evals and working with the Inspect framework by leading researchers in the field.

Data Platform Engineer / Tech Lead / Data Scientist *Gothenburg, Sweden*
Knowit Solutions Cocreate *Feb 2022 – present*

- Mentored junior engineers
- Compared multiple LLM observability tools for use with a job-ad-to-consultant matching LLM platform. (LangChain, Langfuse, Arize Phoenix).
- Deployed Arize Phoenix to monitor LLM calls and model performance in production environments.

Security Analyst *Gothenburg, Sweden*
NTT Security *Apr 2019 – Jan 2022*

- Developed predictive analytics tooling for detecting malicious network traffic, leveraging Python, machine learning, and distributed processing.
- Provided mentorship and security training, educating analysts on threat intelligence, anomaly detection, and alert triage automation.

Python Software Engineer *Gothenburg, Sweden*
Ericsson *June 2017 – Oct 2018*

- Led a team to develop a log parsing and analytics tool that scaled into a dedicated engineering team at Ericsson.

Open Source and Personal Projects

Open Source Contributions
Contributions to data engineering open source projects

- Dagster Contribution: PR #24188 – Enhanced workflow orchestration for ML pipeline monitoring
- DLT Hub Contribution: PR #594 – Improved data ingestion reliability for ML applications

Malicious Domain Prediction System *2020 – 2021*
Real-time ML prediction application for assessing malicious domains for cyber security operations

- Reduced analysis of high-volume bad domains from 1 hour per shift to less than 20 minutes per shift
- Implemented real-time inference pipeline with performance monitoring

Data Breach Analysis Pipeline *2024 – 2025*
Modern data lakehouse solution for analyzing large-scale password breach data

- Developed scalable data processing pipeline using Apache Iceberg, Flink, and Spark to analyze high-volume breach data
- Passion project to test new technologies that required big data

Skills

Machine Learning & AI: PyTorch, Jax, Einops, XGBoost, SciKit Learn, Jupyter, LLM Observability (Inspect, LangChain, Langfuse, Arize Phoenix)

Programming Languages: Python, Javascript, SQL, Rust, C/C++, R, Nix, Terraform

Platforms & Tools: Modal, Runpod, Openweights, Weights & Biases, Spark, Flink, Iceberg, Databricks, Dagster, Snowflake, dbt, Apache Superset

Infrastructure & DevOps: Nix, Docker, Kubernetes, FluxCD, GitOps, CI/CD pipelines, Terraform, AWS (EC2, S3, IAM)

Education

Chalmers Technical University — *Gothenburg, Sweden*
MS in Engineering Physics — *Sept 2016 – June 2018*

- Coursework: Statistical Physics, Neural Networks and Machine Learning

Gothenburg University — *Gothenburg, Sweden*
BS in Computer Science — *Sept 2013 – June 2016*

- Coursework: Algorithms, Testing, Debugging and Verification, Theoretical Computer Science, Operating Systems, Cryptography, Cyber Security